

MEDBOX ALERT: AI-IOT BASED SMART PILL ASSISTANCE FOR INTELLIGENCE HEALTHCARE

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Abstract:

Planning a trip can often feel overwhelming, as it usually requires browsing multiple websites to compare destinations, hotels, routes, weather conditions, and budgets. This research work aims to simplify that process by designing and developing an AI-based Intelligent Trip Planning and Recommendation System. The system is built to help travelers plan their trips in a faster, easier, and more efficient way. By using Artificial Intelligence and Machine Learning, the system can automatically create personalized travel plans based on user preferences such as budget, travel dates, number of travelers, interests, and preferred activities. Instead of manually searching and comparing options, users receive smart recommendations tailored specifically to their needs. To make the system more powerful and practical, it integrates several APIs. Gemini AI is used to generate intelligent suggestions, OpenStreetMap provides mapping and navigation features, a Weather API offers real-time weather updates, and RazorPay ensures secure online payment processing. The system is developed using Python and the Django framework, which supports strong performance, scalability, and data security. Overall, this solution reduces the time and effort involved in trip planning while improving the overall user experience. It makes travel planning more convenient, accurate, and user-friendly.

Keywords: medbox alert, AI-IOT

1. Introduction

Artificial Intelligence has significantly transformed many industries by enabling smarter automation and more personalized services. However, the travel and tourism sector still relies heavily on manual planning. Travelers often need to visit multiple platforms to gather information about destinations, hotels, transportation, and weather conditions. This scattered process can be time-consuming and sometimes overwhelming.

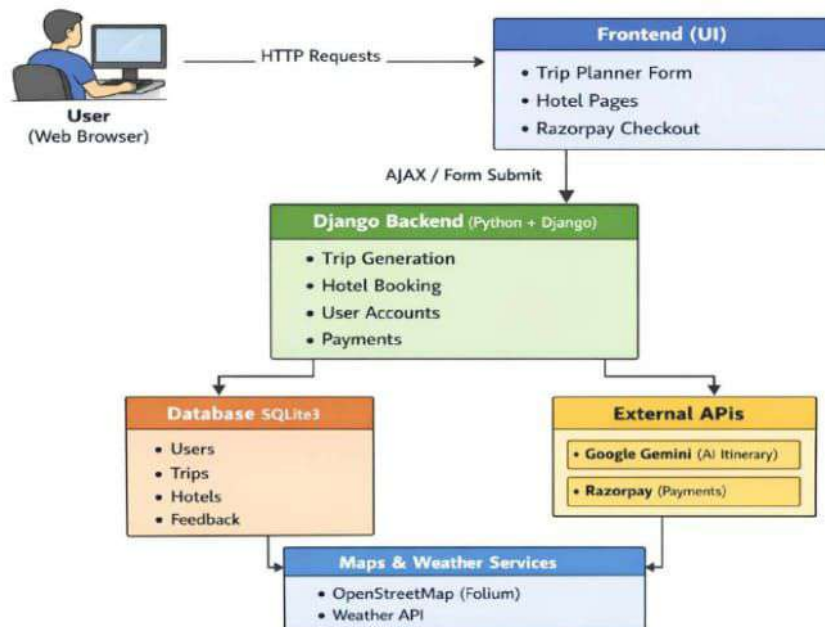
Many people find it difficult to organize all these details into a clear and well-structured travel plan that matches their preferences and budget. To address this issue, this project introduces an AI-based Intelligent Trip Planning and Recommendation System designed to automate the entire planning process.

The system works by analyzing user inputs such as preferences, travel history, and budget constraints to generate personalized travel itineraries. It also offers smart destination suggestions, real-time weather updates, route guidance, and booking support through integrated APIs. By combining multiple travel services into a single platform, the system makes trip planning simpler, reduces the effort required from users, and improves the overall travel experience.

2 Literature Review

- Kumar, A., et al. (2024) developed an AI-based route optimization system for travelers. Their approach used advanced algorithms to identify the most efficient travel routes and predict traffic conditions. The study reported an improvement of nearly 30% in route planning accuracy compared to traditional manual map searches [1].
- Patel, R., et al. (2025) focused on smart tourism by integrating hotel booking with multiple APIs. Their system combined weather, mapping, and payment services into a single platform. The results showed that such integration made online travel booking faster, more convenient, and secure for users [2].
- Zhang, Y., et al. (2023) introduced a chatbot-based tourist assistance system. This chatbot was capable of guiding travelers, suggesting nearby attractions, and answering user queries in real time. Users reported a smoother and more interactive travel experience with this system [3].
- Lee, H., et al. (2024) proposed a deep learning-based system for personalized itinerary generation. By analyzing traveler behavior and preferences, their model was able to create customized travel plans. The study found that users spent less time planning and were more satisfied with AI-generated itineraries [4].
- Singh, P., et al. (2024) developed an AI-driven travel recommendation system using machine learning techniques. Their system analyzed user preferences, travel history, and budget constraints to suggest suitable destinations. The findings showed improved recommendation accuracy and higher user satisfaction, while also reducing manual planning effort [5].
- Verma, S., et al. (2023) proposed a cloud-based intelligent travel planning and booking platform. Their system integrated hotel booking, route mapping, and real-time data services. The study highlighted that cloud technology enhanced system scalability and reduced response time for users [6].
- Chen, L., et al. (2025) designed an AI-powered decision support system for smart tourism. This system helped users select destinations, accommodations, and activities more effectively. Their research indicated that AI-assisted decision-making improved booking efficiency and provided more personalized travel experiences [7].
- Sharma, M., et al. (2023) developed an AI-based hotel recommendation and pricing system. The system suggested hotels based on factors such as budget, location, and user ratings. Results showed that automated recommendations improved booking accuracy, reduced search time, and helped users find affordable accommodations [8].
- Gupta, N., et al. (2024) created an intelligent travel planning system using big data analytics. By analyzing user behavior, reviews, and travel trends, the system generated more relevant travel plans. The study demonstrated that data-driven insights significantly improved personalization in tourism applications [9].

3 Methodology



Data Collection and User Input

The system begins by collecting essential information from users, such as their preferred destination, travel budget, trip duration, and number of travelers. In addition to these inputs, it also gathers external data like weather conditions and location details through integrated APIs. This combination of user input and real-time data helps create more accurate and relevant travel plans.

AI-Based Recommendation Engine

The core of the system is a hybrid recommendation engine that combines rule-based logic with AI algorithms. It processes user inputs to suggest suitable destinations, hotels, and attractions. While generating recommendations, the system considers multiple factors such as popularity, cost, distance, and current weather conditions to ensure practical and personalized results.

System Development Framework

The application is developed using Python along with the Django framework. This combination provides a strong foundation for building a modular, secure, and scalable system. For data storage, SQLite is used to manage user profiles, booking details, and travel history efficiently.

Payment Gateway Integration

To support secure online transactions, the system integrates Razorpay as its payment gateway. This allows users to make bookings for hotels and other travel services safely, with encrypted and reliable payment processing.

Mapping and Location Services

The system uses OpenStreetMap along with the Folium API to provide interactive maps and location-based services. These tools help users visualize travel routes, explore nearby attractions, and identify hotels, making trip planning more precise and user-friendly.

4 Results & Analysis

The developed system successfully provides a range of useful features, including generating customized trip itineraries, offering real-time weather updates, displaying travel routes visually, and enabling hotel bookings. All these features are integrated into a single platform, allowing users to plan their trips smoothly without needing to switch between multiple websites or applications.

Experimental testing showed that the system significantly improves travel planning efficiency compared to traditional methods. Users were able to make decisions much faster due to automated recommendations and the availability of real-time information. This reduced the overall time spent on planning and simplified the entire process.

User feedback further highlighted the effectiveness of the system. Most users reported higher satisfaction levels, noting that the platform made trip planning more convenient, accurate, and reliable. Overall, the system enhances the travel experience by making planning easier and more efficient.

5 Conclusion

The AI Trip Planner and Recommendation System provides an intelligent and automated solution for modern travel management. By integrating AI-driven recommendations, map-based visualization, real-time weather updates, and secure online payments, the system simplifies the entire trip planning process. The use of Django, OpenStreetMap, and Razorpay ensures reliability, scalability, and user convenience. The AI chatbot enhances engagement by assisting users with instant travel queries and suggestions. Overall, the system demonstrates how artificial intelligence and web technologies can transform traditional tourism into a smarter, data-driven, and user-friendly experience. In addition, the inclusion of an AI-based chatbot assists users in real time, providing travel suggestions and addressing queries efficiently. Real-time weather updates and data-driven recommendations further enhance the decision-making process.

6 Future Scope.

The AI Trip Planner and Recommendation System can be further enhanced to make travel planning more advanced, adaptive, and intelligent. Future developments may include the integration of Machine Learning models to analyze user behavior and improve the accuracy of destination and hotel recommendations. The system can also be extended with real-time travel data, such as flight and train schedules, live traffic updates, and dynamic pricing information. Integration with IoT and wearable devices can enable context-aware travel assistance, such as navigation alerts or health-based travel suggestions. Adding multilingual and voice assistant support will make the platform accessible to a wider audience. Additionally, connecting with social media platforms can allow users to share trip experiences and receive peer-based suggestions.

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